

Retail Sales Performance Analysis (USD)

Step-by-step guide for building an active dashboard

This guide explains how to build Project 2 in Power BI Desktop using the prepared USD sales dataset. All currency visuals should be formatted as USD (\$).

Step 1: Load the data

1. Open Power BI Desktop.
2. Go to Home → Get Data → Text/CSV.
3. Load Fact_Sales.csv, Dim_Product.csv, Dim_Customer.csv, Dim_Date.csv, Sales_Targets.csv and Returns.csv.
4. Check that Order_Date, Ship_Date, Date and Return_Date are set as Date type.
5. Check that all columns ending in _USD are set as Decimal Number.

Step 2: Create relationships

From table / column	To table / column	Relationship
Dim_Date[Date]	Fact_Sales[Order_Date]	One-to-many
Dim_Product[Product_ID]	Fact_Sales[Product_ID]	One-to-many
Dim_Customer[Customer_ID]	Fact_Sales[Customer_ID]	One-to-many
Fact_Sales[Order_ID]	Returns[Order_ID]	One-to-many or many-to-one depending on load result

Step 3: Add DAX measures

Create a measure table or place the measures in Fact_Sales. Copy the measures from the supplied file: PowerBI_DAX_Measures_Project2_USD.txt.

Measure	Meaning
Net Sales	Revenue after discount in USD.
Gross Sales	Revenue before discount in USD.
Total Profit	Net sales less total cost in USD.
Profit Margin	Profit divided by net sales.
Orders	Distinct order count.
Average Order Value	Net sales divided by orders.
Return Rate	Returned orders divided by total orders.
Target Achievement	Net sales divided by sales target.

Step 4: Apply USD formatting

1. Select Net Sales, Gross Sales, Total Profit, Average Order Value and Sales Target measures.
2. Go to Measure tools → Format.

3. Set format to Currency and use the dollar symbol (\$).
4. Set Profit Margin, Return Rate and Target Achievement as Percentage with 2 decimal places.
5. Set Orders and Quantity Sold as Whole Number.

Step 5: Build Page 1 – Executive Overview

Add a title: Retail Sales Performance Dashboard (USD).

Visual	Fields / measures	Purpose
KPI cards	Net Sales, Total Profit, Profit Margin, Orders, Average Order Value, Return Rate	Show business performance at a glance.
Line and clustered column chart	Axis: Dim_Date[YearMonth]; Values: Net Sales and Total Profit	Show monthly sales and profitability trend.
Bar chart	Axis: Category; Value: Net Sales	Show top categories by revenue.
Bar chart	Axis: Channel; Value: Net Sales	Compare sales channels.
Bar chart	Axis: Product_Name; Value: Net Sales	Show top products.

Step 6: Add active slicers

1. Add slicers for Year, Category, Segment, Province and Channel.
2. Place slicers across the top of the page.
3. Use dropdown style for Category, Segment, Province and Channel.
4. Use list or dropdown style for Year.
5. Test by selecting a Category and checking that all KPI cards and charts change.

Step 7: Build Page 2 – Sales Trends

Visual	Fields / measures
Monthly trend	Axis: YearMonth; Values: Net Sales and Total Profit
Orders trend	Axis: YearMonth; Value: Orders
Average order value trend	Axis: YearMonth; Value: Average Order Value
Return rate trend	Axis: YearMonth; Value: Return Rate

Step 8: Build Page 3 – Product & Category Analysis

Visual	Fields / measures
Top 10 products	Axis: Product_Name; Value: Net Sales; Visual filter: Top 10 by Net Sales
Category profit margin	Axis: Category; Value: Profit Margin
Subcategory sales	Axis: Subcategory; Value: Net Sales
Product table	Product_Name, Category, Net Sales, Total Profit, Profit Margin, Quantity Sold

Step 9: Build Page 4 – Customer & Region Insights

Visual	Fields / measures
Segment sales	Axis: Segment; Value: Net Sales
Province sales	Axis: Province; Value: Net Sales
Channel by segment	Axis: Channel; Legend: Segment; Value: Net Sales
Customer table	Customer_Name, Segment, Province, Net Sales, Orders

Step 10: Build Page 5 – Insights & Recommendations

Add four text boxes: Key Findings, Business Meaning, Data Limitations and Recommendations. Keep the wording concise and business-focused.

Step 11: Make the dashboard interactive

1. Click a chart.
2. Go to Format → Edit interactions.
3. Choose the filter icon on visuals that should respond.
4. Avoid making all visuals too crowded; keep the table on a separate page if needed.
5. Test each page by selecting a year, category, segment, province and channel.

Step 12: Save and export screenshots

1. Save the file as Rumbidzai_Project2_Retail_Sales_Analysis_Dashboard_USD.pbix.
2. Take screenshots of each page.
3. Save screenshots in the screenshots folder using clear names.
4. Upload the PBIX, screenshots, Excel workbook, SQL script, case study and README to GitHub.